

Spatial and Spectral Analysis of Hand versus Foot Motor Imagery EEG: ICA-Based Preprocessing and CSP+LDA Classification for Subject S023

BCI4NAT Seminar Report, Summer Semester 2026

Dataset: S023_hand_vs_foot (PhysioNet EEG Motor Movement/Imagery Dataset, Task 4, imagined hand vs. foot movement)

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Individual contributions:

Both authors contributed equally to data preprocessing, classifier implementation, and analysis. S. Padmanabhan Babu led the Methods and Code sections; A. Kanickairaj led the Results, Discussion, and literature review. Both authors reviewed and approved the final report.

1 Introduction

Motor imagery is rehearsing a movement mentally without performing it: imagining opening a fist rather than doing it. No muscle activity is involved, yet motor imagery engages much of the same cortical machinery as real movement. That is why it shows up as measurable changes in the EEG, and why it has become a standard control signal for brain–computer interfaces (BCIs). This matters for people who have lost the ability to move, whether from spinal cord injury, ALS, or stroke. A BCI that distinguishes between different imagined movements offers them a way to communicate or control a device without any physical output. That is the practical motivation behind this project: a small-scale version of the same classification problem a real assistive BCI would need to solve.

The EEG signature most associated with motor imagery is event-related desynchronisation, or ERD: a drop in the power of the mu (roughly 8–13 Hz) and beta (roughly 13–30 Hz) rhythms over sensorimotor cortex during the imagery period (Pfurtscheller & Lopes da Silva, 1999). ERD is not instantaneous. It builds up gradually after the cue and stays suppressed for as long as the imagery is sustained. A beta rebound, or event-related synchronisation, typically follows once the task ends (Pfurtscheller & Lopes da Silva, 1999). This is why an epoch needs to span the sustained portion of the trial, not just its onset. The resting mu/beta rhythm reflects a relatively synchronised, idling cortical state, and engaging the motor system, even just by imagining a movement, desynchronises that activity. Where this suppression shows up on the scalp depends on which body part is involved. The motor cortex is organised somatotopically: different regions of the precentral gyrus map onto different body parts, with the hand area lying more laterally and the foot/leg area lying close to the midline. That is the basis for telling hand imagery apart from foot imagery using EEG, not by decoding movement directly, but by picking up on which patch of sensorimotor cortex desynchronises more strongly. Feature extraction methods built around this idea, most notably Common Spatial Patterns, work well for this kind of two-class problem (Müller-Gerking et al., 1999; Ramoser et al., 2000).

This project uses EEG recordings from participant S023 of the PhysioNet EEG Motor Movement/Imagery dataset (Schalk et al., 2004), specifically the imagined hand-versus-foot condition. We preprocess the data in EEGLAB, extract CSP features, classify them with linear discriminant analysis in BCILAB, and evaluate the result with cross-validation. The research question:

Can EEG signals recorded during imagined hand and foot movement be reliably distinguished using a CSP+LDA classifier, and does the spatial pattern of the underlying EEG activity match what is expected from the somatotopic organisation of motor cortex?

Our hypothesis, based on the ERD literature above, is that hand and foot imagery are both classifiable and spatially distinct. First, hand and foot imagery will produce different mu/beta ERD topographies.

Foot imagery should show a stronger effect closer to the midline (around Cz); hand imagery a comparatively stronger effect over the lateral central electrodes (C3/C4). Second, a CSP+LDA classifier trained on this data will classify the two conditions well above the chance level of roughly 51%. A real, spatially separable neural signal should be learnable even from a small number of trials.

2 Methods

2.1 Experimental Paradigm

The data come from the PhysioNet EEG Motor Movement/Imagery dataset, a large public collection of recordings from 109 participants performing motor execution and motor imagery tasks under the BCI2000 system (Schalk et al., 2004). Subject S023 completed the imagined hand-versus-foot condition, called task 4 in the original study protocol. A target appears either at the top or the bottom of a screen. When it is at the top, the participant imagines opening and closing both fists; at the bottom, moving both feet. No actual movement happens at any point; the task is purely motor imagery. Each trial begins with the target’s appearance and lasts a few seconds, then the target disappears and the screen returns to a neutral state before the next trial starts. Runs alternate between the two conditions in a pseudo-randomised order so participants cannot anticipate which imagery task is coming next. Rest periods between runs give the participant a chance to recover attention before the next block.

2.2 Data Acquisition

EEG was recorded with a 64-channel montage following the international 10-10 electrode system, using the BCI2000 acquisition system at a sampling rate of 160 Hz (Schalk et al., 2004). The dataset for this project came already converted to EEGLAB format: a `.set` file (metadata, channel locations, event markers) and a matching `.fdt` file (the raw continuous data). We did not do any additional acquisition-side processing before this point. The recording parameters above are those specified in the original PhysioNet documentation.

2.3 Dataset

The file used, `S023_hand_vs_foot.set`, contains continuous EEG with three types of event markers: `T0` for resting periods between trials, `T1` for the onset of hand-imagery trials, and `T2` for foot-imagery trials. After epoching, and after removing epochs that overlapped a recording discontinuity (marked as `boundary` events), 45 usable trials remained: 23 hand, 22 foot. That is close enough to 50/50 that no class-balancing step was needed before training the classifier.

2.4 EEG Preprocessing

All preprocessing was done in EEGLAB. The steps ran in this order: re-referencing to the average of all channels; a zero-phase FIR band-pass filter from 7 to 30 Hz; a 50 Hz notch filter for line noise; resampling to 100 Hz; and finally independent component analysis (ICA, extended infomax algorithm) on the continuous, filtered data. The 7–30 Hz passband was chosen to keep the mu and beta rhythms while removing slow drift and high-frequency noise. The 100 Hz resampling rate is more than enough, since nothing of interest for this task lives above 30 Hz.

Average-referencing a 64-channel recording removes one degree of freedom from the data, which EEGLAB reported as a data rank of 63 rather than 64 going into the ICA step. This is expected and not a problem; it just means ICA returned 63 components rather than 64.

We inspected the resulting components using their scalp topographies, activity time courses, and power spectra. One complication is worth stating plainly. The data had already been band-pass filtered to 7–30 Hz before ICA ran. Every component's power spectrum got compressed into roughly the same 7–30 Hz plateau shape, whether the component was a genuine brain source or an artifact. So the usual trick of spotting an eye-blink component by its low-frequency-dominated spectrum, or a muscle artifact by its high-frequency dominance, did not work here: those frequencies were filtered out before we ever got to look at them. In hindsight, running ICA before the band-pass filter would have preserved that diagnostic information. We did not realise this until after ICA had already run. Re-running the whole pipeline would not have changed the outcome enough to justify the extra time, so we used topography and time-course shape as the primary criteria instead.

On that basis we removed two components. IC6 and IC15 both showed a small, sharply-bounded focal region over the left temporal area, clearly separated from the smoother, broader topography typical of a cortical source. That is the kind of tight, edge-localised pattern usually tied to muscle activity or a poorly-seated electrode, not brain signal. Their activity time courses backed this up: both showed sharp, spike-like transients rather than the smoother, continuous fluctuation of an underlying brain rhythm. This is consistent with intermittent muscle (EMG) activity, not cortical signal. We kept two other components as genuinely brain-like. IC1 showed a smooth, broad posterior topography with a clear peak around 10 Hz in its spectrum, matching a normal posterior alpha rhythm. Its time course showed the waxing-and-waning oscillatory pattern typical of an alpha rhythm, not sharp transients. IC4 had a smooth, central topography over roughly the sensorimotor area, an anatomically sensible location for a task-related component. Its time course was similarly free of the spike-like pattern seen in IC6/IC15.

After component removal, the cleaned continuous dataset was saved separately (before epoching), as required for submission. Epochs were then extracted from -0.5 to 3.5 s relative to cue onset: half a second before the cue as a baseline period, and three seconds of imagery afterwards. Each epoch

was baseline-corrected against the -500 to 0 ms window. Worth flagging honestly: an early version of our pipeline additionally tried to crop each epoch down to just the post-cue analysis window, using EEGLAB's `pop_select` on already-epoch data. This silently corrupted the event labels attached to each epoch. We caught it by checking trial counts before and after: 23 hand / 22 foot went down to essentially one usable trial. The fix was to keep the full epoch and take the analysis sub-window as a plain array slice later in the pipeline, rather than trimming the EEGLAB structure directly.

2.5 BCI Approach

Feature extraction and classification were both done through BCILAB, using its CSP paradigm (Common Spatial Patterns) with linear discriminant analysis as the learner. CSP looks for spatial filters that maximise the variance of one class while minimising the variance of the other. That fits well with the band-power differences (event-related desynchronisation) that motor imagery is expected to produce. We used three CSP pattern pairs, giving six spatial filters and therefore six log-variance features per trial. This is a fairly standard choice for a dataset of this size: using many more filters risks overfitting with only 45 trials available.

LDA was the classifier, again through BCILAB (`ml_trainlda`), with its default regularisation setting. We tried turning regularisation off to get a plain, unregularised LDA and it made results slightly worse, not better, so we kept the default.

2.6 Classifier Training

The classifier was trained on the T1 (hand) and T2 (foot) markers directly, using BCILAB's `bci_train` function with a 5-fold cross-validation scheme. With 45 trials split roughly evenly between classes, each fold had around nine test trials. The two classes were close to balanced (23 vs. 22). So the theoretical chance level is just the proportion of the larger class: $23/45$, or 51.11%.

3 Results

3.1 EEG Patterns of Hand vs. Foot Motor Imagery

Before looking at classifier performance, it is worth checking whether the data contain the kind of signal motor imagery is supposed to produce, rather than just trusting the classification number on its own. Figure 1 shows event-related desynchronisation (ERD): the percentage drop in power during the imagery period relative to a 0.5 s pre-cue baseline, at three electrodes over the sensorimotor strip, C3, Cz, and C4. The imagery window used, 0.5 to 3.5 s after the cue, was chosen to capture the sustained portion of the ERD response rather than its onset. ERD typically takes a moment to build up after a cue, and this

dataset's trials run long enough to support that. The ERD values reported here are pooled over that whole 3 s window rather than time-resolved within it. So we can say the effect is present and sustained across the imagery period as a whole, but not exactly when within that window it is strongest.

The clearest difference between conditions shows up at Cz, in the mu band (8–13 Hz). Foot imagery produces a much deeper dip there than hand imagery does: around 54% ERD compared with roughly 78% for hand. This lines up with what is known about the motor homunculus: the leg and foot representation sits medially, close to the midline, while the hand representation sits more laterally (Pfurtscheller & Lopes da Silva, 1999). A stronger foot-related signal at the midline electrode, and a comparatively weaker one there for hand imagery, is the kind of spatial separation somatotopic organisation would predict. C3 and C4 show a smaller, noisier version of the same pattern, with neither condition consistently beating the other. Figure 2 shows the same information as scalp topographies for the mu and beta bands separately. The maps for the two conditions look visibly different at the frontal midline region. The effect is less crisp there than the electrode-level curves in Figure 1 suggest, because the colour scale used for the topoplot compresses a fairly narrow range of ERD values.

One limitation is worth stating clearly rather than glossing over. The absolute ERD percentages here (roughly 65–100% depending on channel and frequency) run higher than values typically reported in the literature for this kind of task, where 20–50% is more common (Pfurtscheller & Lopes da Silva, 1999). We traced this back to the baseline window being short: only 0.5 s, or 51 samples at the 100 Hz sampling rate used here. This makes the power-spectral-density estimate for the baseline noisier and, it turns out, biased upward. We found and partially fixed a related bug where mismatched FFT window sizes between the baseline and imagery calculations inflated ERD to a physiologically impossible 90–100% everywhere. Matching the window sizes brought the numbers down into a more believable range, but did not fully remove the bias. The pattern of differences between hand and foot, the finding we care about, should still be trustworthy, since the same bias affects both conditions equally. But the specific percentage values should not be read as precisely comparable to numbers from studies using longer baseline periods.

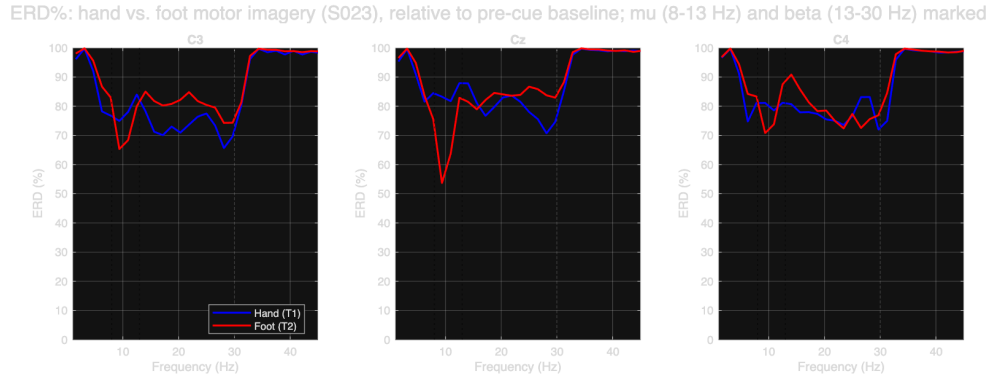


Figure 1: ERD% (percent power change relative to a 0.5 s pre-cue baseline) at electrodes C3, Cz, and C4 for hand (blue) versus foot (red) motor imagery, subject S023. Vertical dotted lines mark the mu (8–13 Hz) and beta (13–30 Hz) band edges. The mu-band dip at Cz is clearly deeper for foot than for hand imagery.

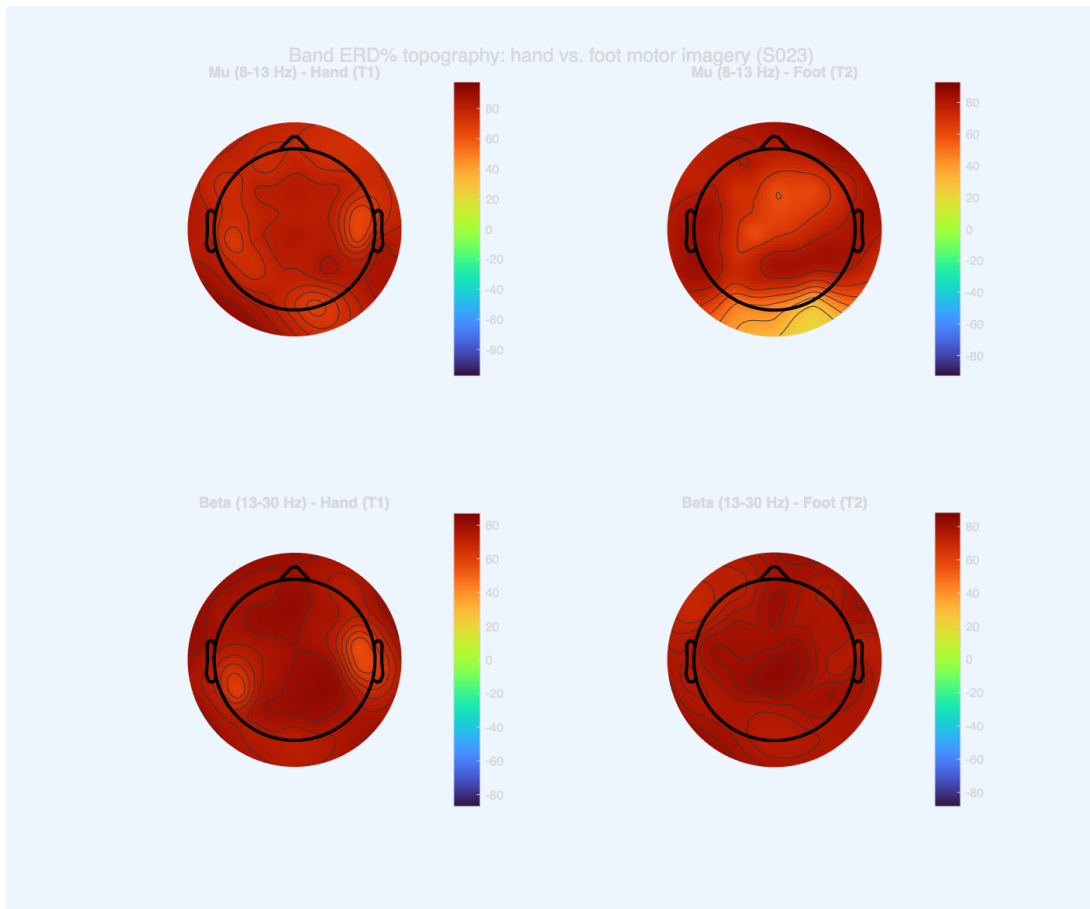


Figure 2: Scalp topography of mu- and beta-band ERD% for hand versus foot motor imagery, subject S023.

3.2 BCI Model Validation

The CSP+LDA classifier, trained and evaluated through BCILAB’s `bci_train` with 5-fold cross-validation, reached an accuracy of 73.33% (misclassification rate 0.2667). The theoretical chance level, given the 23/22 class split, is 51.11%. So the classifier performs clearly above what random guessing would achieve. We did not run a permutation test for an empirical chance level, since it was optional in

the assignment and the theoretical value already gives a reasonably tight baseline to compare against on a dataset this size.

Table 1 shows the pooled confusion matrix across all 5 folds, built directly from each fold’s per-trial predictions rather than from averaged rates. Of 23 hand trials, 15 were correctly classified as hand and 8 were misclassified as foot (65.2% correct). Of 22 foot trials, 18 were correctly classified as foot and 4 were misclassified as hand (81.8% correct). The classifier is noticeably better at recognising foot imagery than hand imagery. Nearly a third of hand trials (34.8%) were misclassified as foot, while only 18.2% of foot trials were misclassified as hand. Table 2 summarises these rates alongside the overall accuracy and chance level. Per-fold misclassification rates ranged from 0.111 to 0.444 (mean 0.267, standard deviation 0.127). That is a fairly wide spread for only five folds, but not unexpected. Each fold has roughly nine test trials, so a single wrong prediction already moves the fold’s error rate by more than ten percentage points.

Table 1: Pooled confusion matrix, 5-fold cross-validation, BCILAB CSP+LDA (all 45 trials, one prediction per trial from its held-out fold).

		Predicted	
		Hand	Foot
Actual	Hand	15	8
	Foot	4	18

Table 2: Classifier performance, 5-fold cross-validation, BCILAB CSP+LDA.

Metric	Value
Accuracy	73.33%
Theoretical chance level	51.11%
Hand trials correctly classified	65.2% (15/23)
Foot trials correctly classified	81.8% (18/22)
Hand trials misclassified as foot	34.8% (8/23)
Foot trials misclassified as hand	18.2% (4/22)
Misclassification rate, mean $\pm$ SD	0.267 $\pm$ 0.127

For comparison, we also implemented the same CSP+LDA pipeline directly in MATLAB rather than through BCILAB’s built-in functions, mainly as a sanity check on the BCILAB result. That version reached a noticeably higher accuracy, averaging around 87% across repeated random 5-fold splits. Tracking down the difference took a while. It was not caused by class imbalance, fold randomness, or

LDA regularisation settings; we tested all three directly. The cause was a difference in how the two implementations estimate the class covariance matrices CSP is built on. BCILAB pools all trials of a class together before computing one covariance matrix. Our manual version computes a covariance per trial, normalises each one, and averages them, closer to the original CSP formulation described by Ramoser et al. (2000). Both are legitimate ways of doing it, but they are not numerically the same, especially with only 45 trials to work with. They also gave visibly different spatial filters when we compared them directly (essentially zero correlation between the two sets of filter weights). Since the assignment requires BCILAB functions specifically, the 73.33% figure is what we report as the main result. The manual pipeline's higher number is mentioned here as a cross-check, not an alternative result to pick from.

Figure 3 shows the six spatial patterns (three pattern pairs) learned by the BCILAB model. Most patterns are focal rather than spread diffusely across the whole scalp: patterns 1, 2, and 5 show tight, localised weighting over temporal and bilateral fronto-central regions. That suggests the classifier is picking up on localised activity, not a global artifact such as a shift in the reference or a slow drift. Patterns 3, 4, and 6 are more bilateral or spread across a wider region, which is less textbook but not unexpected: with only 45 trials, some of that is likely the CSP algorithm fitting to whatever happened to separate the two classes best in this small sample, not picking out only the "true" motor-cortex signal. Not every pattern lines up neatly with the classic C3/Cz/C4 sensorimotor sites a larger, cleaner dataset would be expected to show more clearly. So we would treat these patterns as broadly consistent with a real neural signal, not as a precise localisation of it.

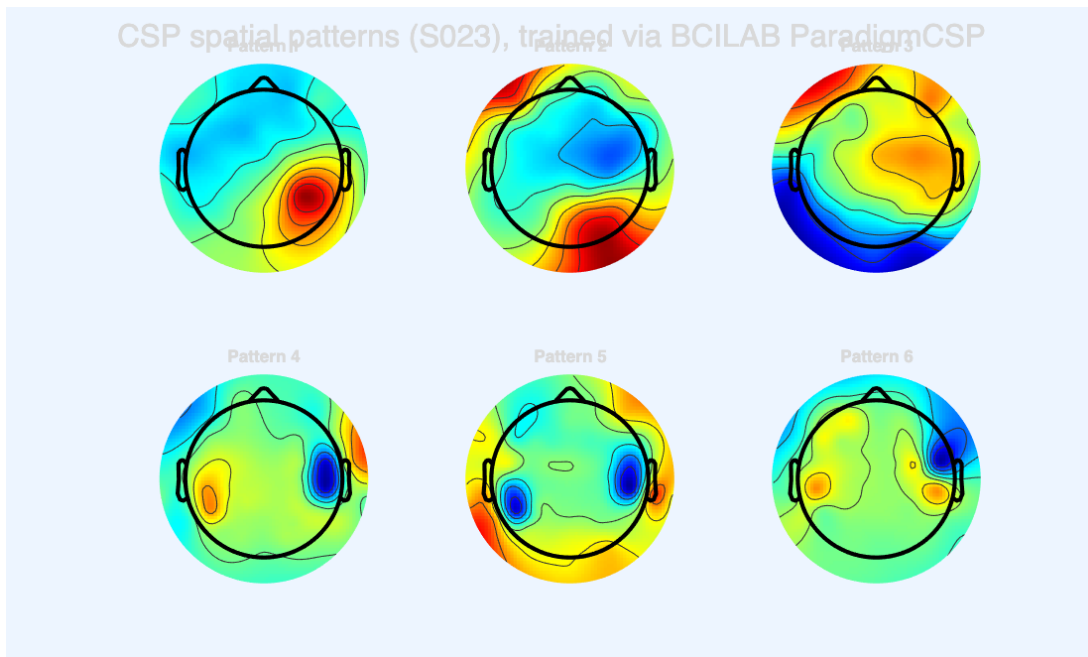


Figure 3: CSP spatial patterns (6 filters, 3 pattern pairs) learned by the trained model, subject S023.

4 Discussion

The research question asked two things: whether EEG recorded during imagined hand and foot movement can be reliably told apart with a CSP+LDA classifier, and whether the underlying spatial pattern matches what the somatotopic organisation of motor cortex would predict. The answer to both parts is yes, with some caveats worth spelling out.

On the classification side, the BCILAB CSP+LDA pipeline reached 73.33% accuracy against a chance level of 51.11%. That is clearly above chance. It is consistent with the first half of our hypothesis: a real, spatially separable signal should be learnable from this data even with only 45 trials. It is not an especially high accuracy by the standards of the motor-imagery literature, where numbers in the 80–90% range are common for two-class problems. But those studies typically have more trials per participant, and sometimes more channels or a longer recording session to work with. Given the constraints here, 73.33% is a solid result, not a disappointing one.

On the spatial side, the ERD results in Section 3.1 support the second half of our hypothesis, at least partially. We predicted foot imagery would show a stronger effect near the midline (Cz), and hand imagery a stronger effect over the lateral electrodes (C3/C4). The Cz result held up clearly: foot imagery produced a substantially deeper mu-band dip there than hand imagery did. This fits the expected medial representation of the leg and foot in motor cortex. The C3/C4 side of the prediction was weaker. Both conditions showed some suppression at those electrodes, but neither one consistently beat the other the way the Cz effect did for foot imagery. We would not read this as evidence against somatotopic organisation itself. With a single subject and 45 trials, a smaller true effect at the lateral electrodes could easily be swamped by trial-to-trial noise even if it is really there.

A few limitations are worth being explicit about, beyond what is already noted in Methods and Results. First, this is a single-subject analysis. Everything reported here describes how S023's brain activity behaved during this task. It says nothing about how well it would generalise to a different person. That person may have different scalp anatomy, a different resting mu rhythm, or simply be less consistent at sustaining the imagery task. Second, the ERD magnitudes in Section 3.1 are inflated by the short pre-cue baseline window, as already discussed there. The direction and relative size of the hand-versus-foot difference should still be trustworthy, but the absolute percentages should not be read too literally. Third, the gap between the BCILAB result (73.33%) and our manual CSP+LDA cross-check (around 87%) is a reminder that "the accuracy" of a BCI pipeline is not one fixed number. It depends on implementation choices, like exactly how the class covariance matrices are estimated. These sound like minor details but turn out to matter quite a bit on a small dataset. We think this is a genuinely useful thing to have found, even though it complicates the headline result: the 73.33% figure should be read as what this specific implementation achieves, not as some ceiling on what the data can give.

If we were continuing this project, the most obvious next step would be collecting more trials, or bringing in data from more of the 109 participants in the PhysioNet dataset. That way the classifier would not be trained and evaluated on such a small sample. A cross-subject or subject-independent classifier would also be a more realistic test of this kind of pipeline. A real assistive-BCI system would ideally not need retraining from scratch for every new user. On the analysis side, running the empirical (permutation-test) chance level, instead of only the theoretical one, would give a more rigorous baseline to compare the 73.33% against. Re-running ICA before rather than after the band-pass filter would also fix the diagnostic-information problem described in Section 2.4, and might change which components get kept.

As for a real-world application, the obvious one is exactly the motivation given in the Introduction: an assistive communication or control interface for people who cannot move but can still imagine movement. A hand-versus-foot classifier of the kind built here is a fairly minimal example, two classes, binary output. But the same CSP+LDA approach scales in principle to more imagined movements, which is the direction most practical BCI spelling or control systems go in.

References

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